Traces of the Past: How Early Language Exposure Affects *Wh*-question Processing

Previous work has shown that long distance (LD) *wh*-extraction is hard for typically-developing children who are learning English as an L1,[1,2,3] children with specific language impairment,[4] and late learners of English as a Second Language (ESL).[5,6] Some *wh*-questions are syntactically ambiguous. For example, the English question *When did the analyst admit he falsified the sales?* can involve short distance (SD) extraction (when the admitting occurred) or LD extraction (when the falsifying occurred). Children who are native English (NE) speakers prefer SD extraction more than NE-speaking adults.[1] We investigated whether very early linguistic background affects how adults process ambiguous *wh*-questions.

Participants. Participants were 83 college students who said English was their preferred written and spoken language. Thirty-three were NE speakers with 2 NE-speaking parents and 50 had no NE-speaking parents. In this paper, participants with no NE-speaking parents will be referred to as Early ESL speakers even though, by self-report, 19 learned only English from birth, 26 learned English and another language from birth, and 5 learned English in early childhood, but said it wasn't their L1. No participants reported a hearing, written, or spoken language impairment.

Procedure. SD versus LD preference was assessed via an online self-paced, whole-sentence reading task. Participants first read a context sentence followed by an ambiguous *wh*-question about the context sentence. Finally, they selected as many answers as they wanted from among 4 options (correct SD and LD answers, and incorrect SD and LD distractors, see Example).

Stimuli. There were 25 *when* questions and 25 *where* questions, each of which appeared five times with the matrix verbs *admit, announce, confess, remember,* and *report*. Stimuli were designed to eliminate confounds such as length, tense, prepositions, gender, etc.. The mean length of context sentences was 14.9 words, and the mean length of questions was 10.2 words.

Results. Responses were considered correct if *only* correct answers were chosen and incorrect if either or both incorrect answers were chosen. A Bayesian ANOVA revealed that NE and Early ESL speakers were equally accurate at interpreting questions ($BF_{10} = 0.042$, Fig 1). NE and Early ESL speakers were also equally fast at reading context sentences ($BF_{10} = 0.08$, Fig 2), and selecting their first answer ($BF_{10} = 0.132$, Fig 3). However, NE speakers were 17.7% faster at reading questions ($BF_{10} = 4.52 \times 10^7$, Fig 4). For questions that were correctly answered, again, NE and Early ESL speakers were equally fast at reading context sentences ($BF_{10} = 0.042$, Fig 5) and choosing their first option ($BF_{10} = 4.30 \times 10^6$, Fig 6), but NE speakers were definitively faster (16.6%) reading the ambiguous questions ($BF_{10} = 4.30 \times 10^6$, Fig 7). Because Early ESL and NE speakers' RTs were the same for reading context sentences (that were 5 words longer than the questions) and choosing answers, Early ESL speakers' longer RTs for ambiguous questions likely reflects their greater difficulty processing ambiguous questions and not them just being slower readers. We next used the Plackett-Luce model to analyze participants' SD vs. LD preference. The Plackett-Luce model considers not only if an option is chosen, but the *order* in which it is chosen if both options are selected. Early ESL speakers preferred the SD option significantly more than the LD option ($\beta = +0.42$, $SE = 0.10$, $p < .001$). NE speakers also tended to prefer the SD option ($\beta = +0.24$, $SE = 0.13$, $p = .054$), but much less so than Early ESL speakers.

Discussion. Our findings that Early ESL speakers process ambiguous questions more slowly and have a greater preference for SD extraction than *true* NE speakers provide novel support for an old proposition: LD *wh*-extraction is hard. Researchers have suggested that LD extraction is difficult because of its computational/processing load,[7] syntactic complexity,[8,9] or a combination.[2] Our findings are most consistent with accounts that include both processing load and syntactic complexity because, although Early ESL participants *preferred* SD extraction, they sometimes extracted LD. These results run contrary to the oft-made claim that adults who are immersed in an L2 by puberty process written syntax like native speakers.[10] In fact, even for adults – who learned English as infants or young children and said English is their dominant language – traces of their early exposure to non-native English or another language linger into adulthood.

Example of an Experimental Trial

Context sentence: *The analyst admitted in January that he falsified the sales last year*
Ambiguous question: *When did the analyst admit he falsified the sales?*
Options: Correct SD (*in January*), Correct LD (*last year*), Incorrect SD (*in July*), Incorrect LD (*last quarter*)

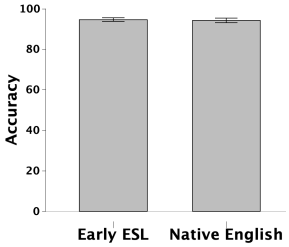

Fig 1. Accuracy (For all figures, error bars are 95% credible intervals)

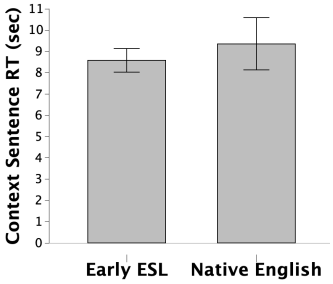

Fig 2. Context Sentence RT (all trials)

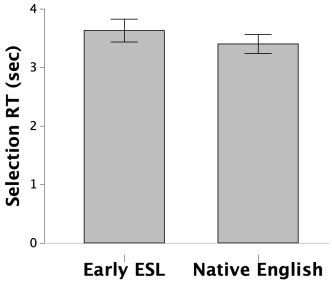

Fig 3. First Selection RT (all trials)

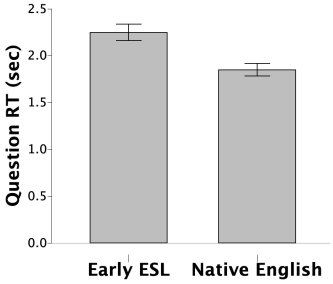

Fig 4. Question RT (all trials)

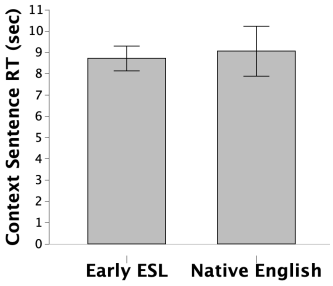

Fig 5. Context Sentence RT (correct trials)

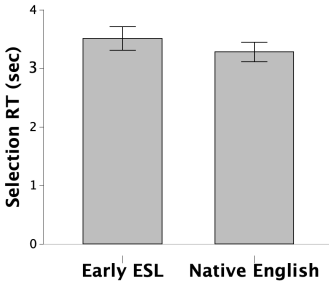

Fig 6. First Selection RT (correct trials)

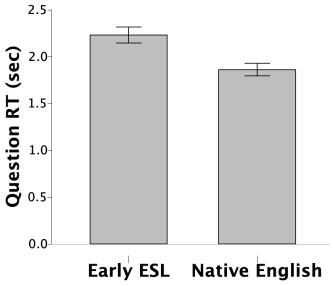

Fig 7. Question RT (correct trials)

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